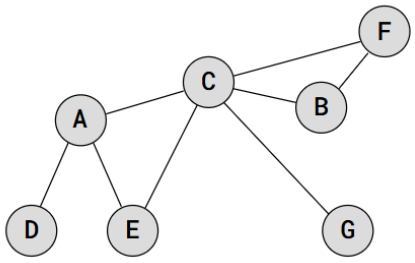
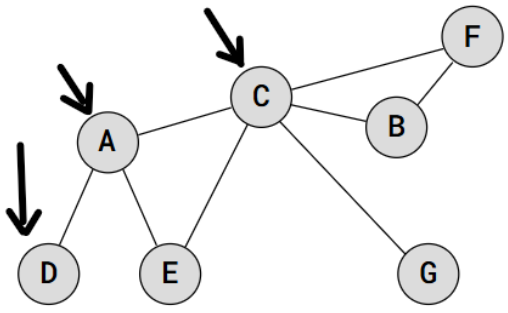
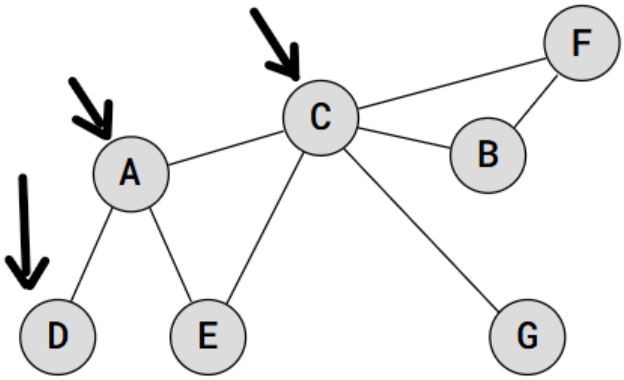
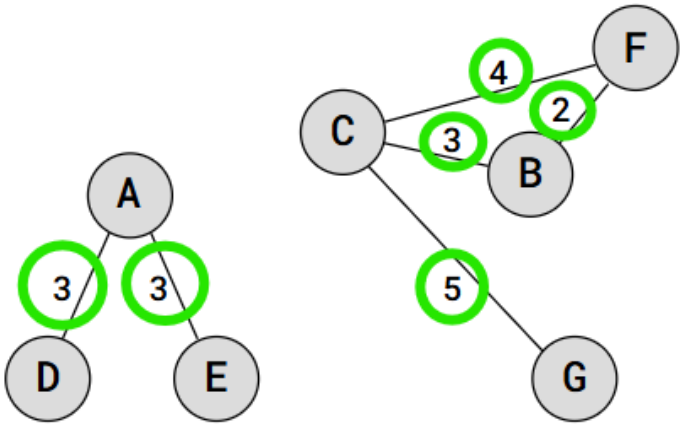
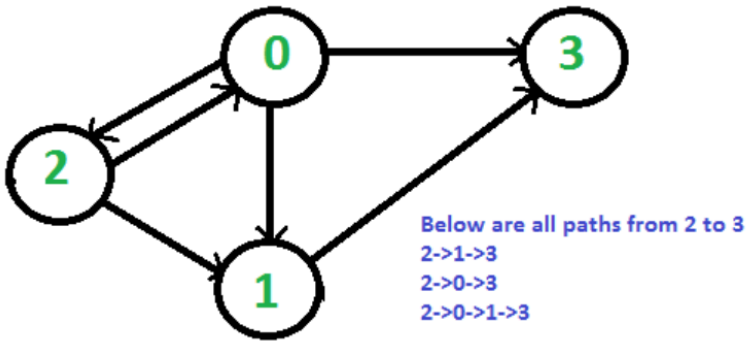
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QUEZON CITY

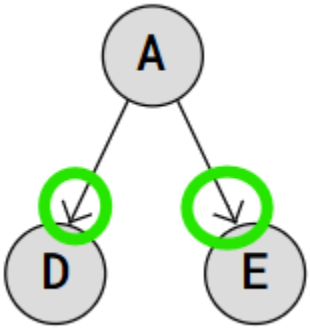
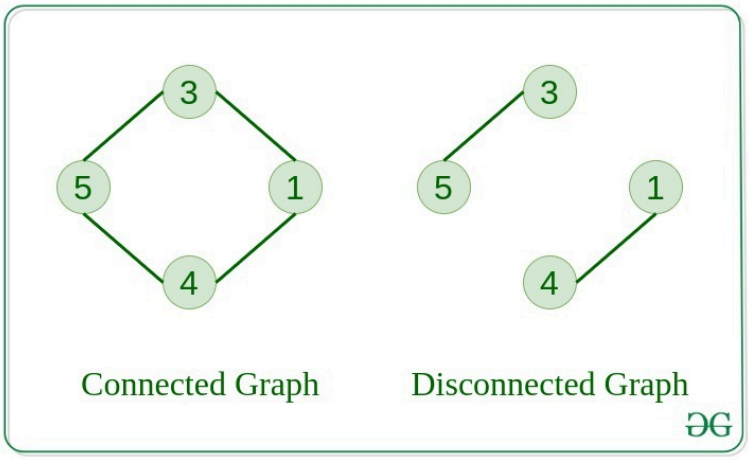
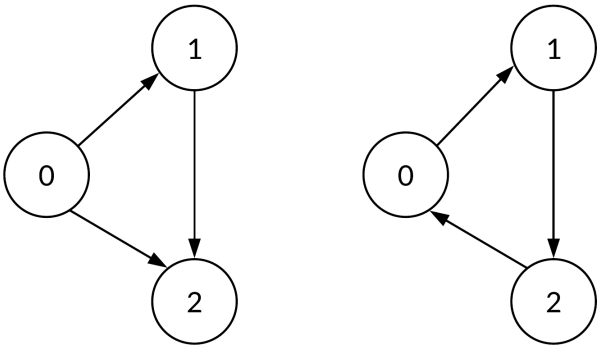
COURSE CODE/TITLE:	CPE 010 - Data Structure & Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	Oct 1, 2026	**DATE SUBMITTED:**	Oct 1, 2026
NAME:	Braganza, Ralph Angelov Bal, Kevin Ranz Llaneta, Jhunel Uchi, Mao Sophie Verano, James Daniel	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 10.1 Graph

1. PROCEDURE OUTPUT

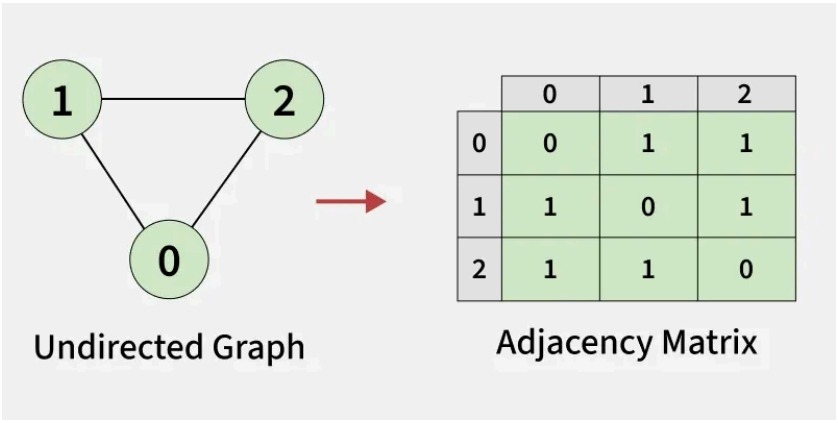
1. Define the following: (10 pts)	
A. Graph	A non-linear data structure consisting of vertices (nodes) connected by edges. It represents relationships between different elements. 
B. Vertex	Also called a node, is a single point or element in a graph that represents an object or entity. 

C. Nodes	The individual elements or points in a graph. They are also called a vertex. 
D. Weight	A numerical value assigned to an edge that represents the cost, distance, or time between two vertices. 
E. Path	A sequence of vertices connected by edges, showing a route from one vertex to another. 

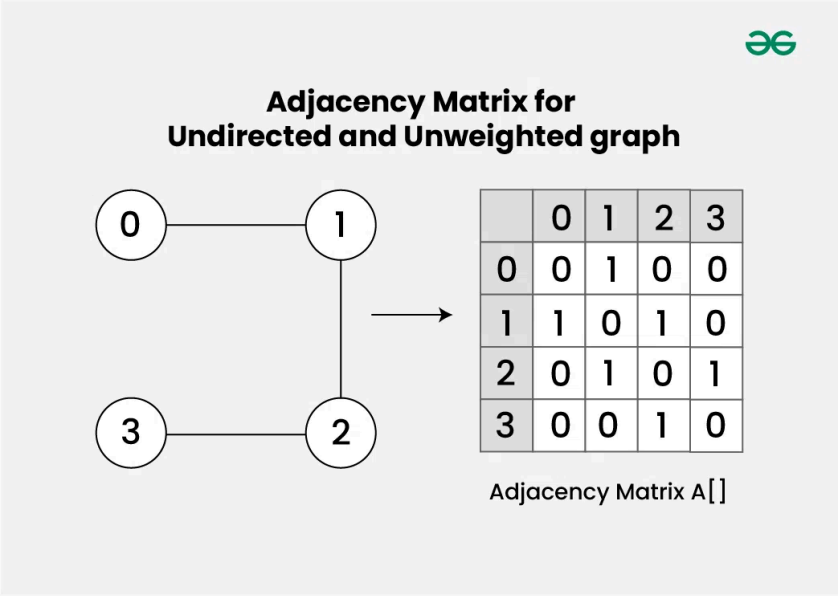
F. Directed graph	A graph where edges have specific directions(indicated with arrows), meaning connections go from one vertex to another in a particular direction. 
G. Connected graph	A graph in which there is a path between every pair of vertices. 
H. Directed cyclic	A directed graph that contains at least one cycle, meaning it is possible to follow the directed edges and return to the starting vertex. Acyclic Graph Cyclic Graph 

I. Adjacency matrix

A two-dimensional array used to represent a graph, where each cell indicates whether two vertices are connected, or stores the weight of their connecting edge.



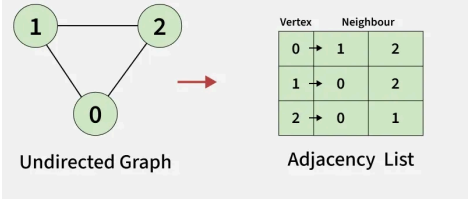
	0	1	2
0	0	1	1
1	1	0	1
2	1	1	0

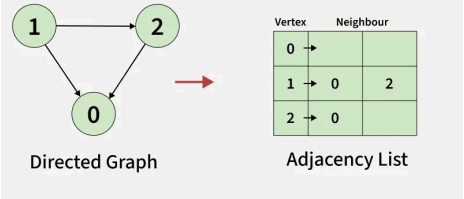


	0	1	2	3
0	0	1	0	0
1	1	0	1	0
2	0	1	0	1
3	0	0	1	0

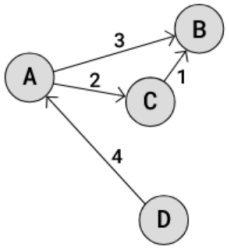
J. Adjacency List

A graph representation that stores each vertex along with a list of its neighboring vertices.





2. Identify the parts of the following graph. (10 pts)



ANSWER:

Vertex	Edge	Weight
V = {A, B, C, D}	Edge A(1) = {A to B} Edge A(2) = {A to C} Edge C = {C to B} Edge D = {D to A} E = {(A, B), (A, C), (C, B), (D, A)}	Weight of {A to B} is 3 Weight of {A to C} is 2 Weight of {C to B} is 1 Weight of {D to A} is 4 W = {(A, B, 3), (A, C, 2), (C, B, 1), (D, A, 4)}

Path (Vertices)	Path (Edges)
Path [A to B] = {A, B} Path [A to B] = {A, C, B} Path [C to B] = {C, B} Path [D to A] = {D, A} Path [D to C] = {D, A, C} Path [D to B] = {D, A, B} Path [D to B] = {D, A, C, B}	Path [A to B] = {(A, B)} Path [A to B] = {(A, C), (C, B)} Path [C to B] = {(C, B)} Path [D to A] = {(D, A)} Path [D to C] = {(D, A), (A, C)} Path [D to B] = {(D, A), (A, B)} Path [D to B] = {(D, A), (A, C), (C, B)}

Path (Weights)	Cycle
Path [A to B] = {(A, B, 3)} Path [A to B] = {(A, C, 2), (C, B, 1)} Path [C to B] = {(C, B, 1)} Path [D to A] = {(D, A, 4)} Path [D to C] = {(D, A, 4), (A, C, 2)} Path [D to B] = {(D, A, 4), (A, B, 3)} Path [D to B] = {(D, A, 4), (A, C, 2), (C, B, 1)}	This directed graph with no cycles is called Directed Acyclic Graph or DAG.

3. Plot the adjacency matrix of the graph above. (10 pts)

	A	B	C	D
A	0	3	2	0
B	0	0	0	0
C	0	1	0	0
D	4	0	0	0

4. Create a program in C++ program to create a graph that looks like in the above figure.

Code

```cpp
int main() {
    // 4x4 matrix initialized to 0
    int matrix[4][4] = {0};

    // vertex variables
    const int A = 0, B = 1, C = 2, D = 3;

    // Set graph edges and weights
    matrix[A][B] = 3;
    matrix[A][C] = 2;
    matrix[C][B] = 1;
    matrix[D][A] = 4;

    std::cout << "  -----------------" << std::endl;
    std::cout << "  | A | B | C | D |" << std::endl;
    std::cout << "  -----------------" << std::endl;

    char labels[4] = {'A', 'B', 'C', 'D'};

    for (int i = 0; i < 4; i++) {
        std::cout << labels[i] << " | ";
        for (int j = 0; j < 4; j++) {
            std::cout << matrix[i][j] << " | ";
        }
        std::cout << std::endl;
        std::cout << "  -----------------" << std::endl;
    }

    return 0;
}
```

Output

```
  -----------------
  | A | B | C | D |
  -----------------
A | 0 | 3 | 2 | 0 |
  -----------------
B | 0 | 0 | 0 | 0 |
  -----------------
C | 0 | 1 | 0 | 0 |
  -----------------
D | 4 | 0 | 0 | 0 |
  -----------------
```

5. Differentiate BFS vs DFS types of search.

The difference between the BFS vs DFS is that BFS **searches level by level** while DFS **searches as far as possible along one path before going back to the previous node and exploring another path**, BFS **Uses queue** while DFS **uses stack or recursion**, BFS **finds the shortest path in an unweighted graph** while DFS **does not always find the shortest path from the node to another**, Usually BFS **needs more memory** than DFS, and lastly BFS is **good for finding nearby or shortest solutions** while DFS **checks all the possible paths of nodes.**

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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